

Problem 1

(a) Shown in class the basis for the product topology of $\mathbb{R}^\omega$ are of the form

$$U_1 \times U_2 \times \cdots \times U_n \times \mathbb{R} \times \mathbb{R} \times \cdots$$

s.t. U_i are open sets in $\mathbb{R}$. Let $x \in \mathbb{R}^\omega$. Let U be an open neighborhood of x . It suffices to show for just a basis set, so let U just be a basis element.

Define a sequence $y = (y_1, y_2, \dots)$ s.t.

$$y_i = \begin{cases} x_i, & \text{if } 1 \leq i \leq n, \\ 0, & \text{if } i > n. \end{cases}$$

Thus, $y \in U \cap \mathbb{R}^\infty$

Thus, every open neighborhood of every point $x \in \mathbb{R}^\omega$ intersects $\mathbb{R}^\infty$. Thus $\overline{\mathbb{R}^\infty} = \mathbb{R}^\omega$ in the product topology.

(b) Let $x = (x_1, x_2, \dots) \notin \mathbb{R}^\infty$ this means that there is no $n \in \mathbb{N}$ s.t. it is all zeros after. Thus consider the neighborhood of x

$$U = \prod_{i=0}^{\infty} U_i$$

s.t.

$$U_i = \begin{cases} (-1, 1), & \text{if } x_i = 0, \\ (x_i - \epsilon_i, x_i + \epsilon_i), & \text{if } x_i \neq 0. \end{cases}$$

$\epsilon_i = |x_i|/2$. This makes sure that the interval does not intersect zero.

This is open since it is made up of open intervals of $\mathbb{R}$.

Let $y \in U$. By definition of U , for each n we have $y_n \in U_n$.

If $x_n \neq 0$ then $y_n \neq 0$, and since there is no n where it ends it infinitely many zeros this means y can't also have infinitely many zeros. Thus $y \notin \mathbb{R}^\infty$

Thus, $U \cap \mathbb{R}^\infty = \emptyset$.

Thus, $\overline{\mathbb{R}^\infty} = \mathbb{R}^\omega$, since any point outside we can create a neighborhood that does not intersect $\mathbb{R}^\infty$

(c) Let $C = \{x = (x_1, x_2, \dots) \in \mathbb{R}^\omega \mid (x_n) \rightarrow 0\}$

Let $x \notin C$ this means x_n does not converge to 0. Thus, there is a $\epsilon_0 > 0$ s.t. for all

$N \in \mathbb{N}$ there is some $n \geq N$ with

$$d(x_n, 0) \geq \epsilon_0$$

Fix $\epsilon = \min\{1, \epsilon_0/2\}$. Consider the open ball

$$B(x, \epsilon) = \{y \in \mathbb{R}^\omega \mid d(x, y) < \epsilon\}$$

since $\epsilon < 1$ we get that

$$B(x, \epsilon) = \{y \in \mathbb{R}^\omega \mid d(x_n, y_n) < \epsilon \text{ for all } n \in \mathbb{N}\}$$

Assume for contradiction that $y \in B(x, \epsilon) \cap \mathbb{R}^\infty$

Fix $M \in \mathbb{N}$ s.t. $y_n = 0$ for all $n \geq M$.

However, since $x \notin C$ we have that $d(x_n, 0) \geq \epsilon_0$ for infinitely many $n \in \mathbb{N}$ thus pick index $n_0 \geq M$ s.t.

$$d(x_{n_0}, 0) \geq \epsilon_0$$

Then since $y_{n_0} = 0$ we have

$$d(x_{n_0}, y_{n_0}) = d(x_{n_0}, 0) \geq \epsilon_0 > \epsilon$$

Thus $y \notin B(x, \epsilon)$ which is a contradiction therefore $x \notin \overline{\mathbb{R}^\infty}$, since there is a neighborhood of x that does not intersect $\mathbb{R}^\infty$

Let $x \in C$ thus $(x_n) \rightarrow 0$

Let U be an open neighborhood of x . Thus there exists an $\epsilon > 0$ s.t.

$$B(x, \epsilon) \subseteq U$$

If $\epsilon > 1$ just fix $0 < \delta < 1$ s.t. $B(x, \delta) \subseteq B(x, \epsilon) \subseteq U$, so WLOG let $\epsilon < 1$.

Since $(x_n) \rightarrow 0$ there is an $N \in \mathbb{N}$ s.t. for all $n \geq N$ we have

$$d(x_n, 0) < \epsilon/2$$

Define a sequence $y = (y_1, y_2, \dots)$ s.t.

$$y_n = \begin{cases} x_n, & n < N, \\ 0, & n \geq N. \end{cases}$$

For $n < N$ we have that $y_n = x_n$ so $d(x_n, y_n) = 0 < \epsilon$

For $n \geq N$ we have that $y_n = 0$ and since $d(x_n, 0) < \epsilon/2$ we have that

$$d(x_n, y_n) = d(x_n, 0) < \epsilon/2 < \epsilon$$

Thus for all $n \in \mathbb{N}$ we have $d(x_n, y_n) < \epsilon$ so

$$d(x, y) = \sup_{n \in \mathbb{N}} d(x_n, y_n) < \epsilon$$

Thus $y \in B(x, \epsilon) \subseteq U$

Thus $x \in \overline{\mathbb{R}^\infty}$

Thus $\overline{\mathbb{R}^\infty} = C$

Problem 2

(a) Assume f_n converges uniformly to f . Let O be open in Y . Consider $f^{-1}(O)$.

Let $z \in f^{-1}(O)$. Thus $f(z) \in O$. Thus there exists some $\epsilon > 0$ s.t.

$$B(f(z), \epsilon) \subseteq O$$

By definition of uniform convergence there is an $N \in \mathbb{N}$ s.t. for $n \geq N$ we have

$$d(f_n(x), f(x)) < \epsilon/3$$

for all $x \in X$

Since $f_N(x)$ is continuous at z there exists an open neighborhood U of z s.t. for all $x \in U$

$$d(f_N(z), f_N(x)) < \epsilon/3$$

Thus, for every $x \in U$, we have

$$d(f(x), f(z)) \leq d(f(x), f_N(x)) + d(f_N(x), f_N(z)) + d(f_N(z), f(z)) < \epsilon/3 + \epsilon/3 + \epsilon/3 = \epsilon$$

Thus for every $x \in U$, we have

$$f(x) \in B(f(z), \epsilon) \subseteq O$$

Thus $U \subseteq f^{-1}(O)$

Thus for any $z \in f^{-1}(O)$ there exists an open neighborhood U in X s.t. $U \subseteq f^{-1}(O)$. This shows that $f^{-1}(O)$ is open. Thus f is continuous.

(b) Let $f_n(x) = x^n$ on $[0, 1] \rightarrow \mathbb{R}$.

If $0 \leq x < 1$

If $x = 0$ then let O be an open neighborhood of 0 in $\mathbb{R}$. Thus there is an $\epsilon > 0$ s.t.

$$B(0, \epsilon) \subseteq O$$

Since $x^n \geq 0$ for all n on the interval $[0, 1]$ it is enough to show that $x^n < \epsilon$ since $0^n < \epsilon$ we are done for that case now just show for $0 < x < 1$

Fix $N > \ln(\epsilon) / \ln(\epsilon x)$, this is well defined since $0 < x < 1$ thus for all $n \geq N$ we get that

$n > \ln(\epsilon) / \ln(\epsilon x) \implies n \ln(x) = \ln(\epsilon) \implies x^n < \epsilon \implies x \in (-\epsilon, \epsilon) \subseteq O$, thus $f_n(x) \rightarrow 0$ for $0 \leq x < 1$

If $x = 1$, since $f_n(1) = 1^n = 1$ it is always in a neighborhood of 1, so $f_n(1) \rightarrow 1$

thus $f_n(x) \rightarrow f(x)$ s.t

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1, \\ 1 & \text{if } x = 1 \end{cases}$$

This is not continuous consider $(0,1)$ open in $\mathbb{R}$. $f^{-1}(0,1) = \{0\}$. Since $[0,1]$ is a subspace of a Hausdorff space we know it is Hausdorff thus singletons are closed, thus $f(x)$ is not continuous.

Problem 3

Separate Proof:

Consider $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ s.t. $f(x) = x + a$ for some $a \in \mathbb{R}^n$. This is continuous since adding is continuous and a is a constant function which is continuous. Consider the function $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ s.t. $g(x) = x - a$. This is an inverse for f and same reason it is continuous. Thus, this is a homeomorphism.

Let $x \in \mathbb{R}^n$ s.t. $n \geq 2$. Now by the proof above we showed that translation is a homeomorphism thus WLOG let $x = 0$.

Consider $\mathbb{R}^n \setminus \{0\}$

Consider the straight-line path $l(x) = (1 - x)a + xb$ this is continuous since we are just adding and multiplying. Also this is a path since $l(0) = a$ and $l(1) = b$.

If 0 is not in our image then we are good, and we get a continuous path from a, b in $\mathbb{R}^n \setminus \{0\}$

If 0 is in our image then pick any point $c \in \mathbb{R}^n$ s.t. there is no line that connects a, b, c . This is possible since $n \geq 2$.

Now consider the two line paths

$$l_1(x) = (1 - x)a + xc$$

$$l_2(x) = (1 - x)c + xb$$

for $x \in [0, 1]$

Now set $l(x)$ to be

$$l(x) = \begin{cases} l_1(2x) & \text{if } (0 \leq x \leq 1/2), \\ l_2(2x - 1) & \text{if } (1/2 \leq x \leq 1), \end{cases}$$

Using the pasting lemma we can see it is continuous on $[0, 1/2]$ and $[1/2, 1]$, and for the intersection since the two functions agree we know that l is continuous.

Thus this is a path in $\mathbb{R}^n \setminus \{0\}$

Thus $\mathbb{R}^n \setminus \{0\}$ is path connected, and therefore connected. Thus $\mathbb{R}^n \setminus \{x\}$ is connected for all $x \in \mathbb{R}^n$

Problem 4

Separate Proof:

Let $f : X \rightarrow Y$ be a homeomorphism. Let $A \subseteq X$ then consider $f|_A : A \rightarrow f(A)$.

Since f is continuous shown in class its restriction is continuous. Now since f is a homeomorphism its inverse is also a continuous bijection thus we can take its restriction map on $f(A)$ thus this is also continuous, and an inverse for $f|_A$.

Thus, $f|_A$ is a homeomorphism.

(a) 1. $(0,1)$ vs. $(0,1]$

Assume: There exists a homeomorphism

$$f : (0,1) \rightarrow (0,1].$$

Since 1 is an element of $(0,1]$, there must be some $x_0 \in (0,1)$ with

$$f(x_0) = 1.$$

- In $(0,1]$, remove 1:

$$(0,1] \setminus \{1\} = (0,1).$$

This set is connected, since we can define $f' : [0,1] \rightarrow (0,1)$ by $f(x) = (1-x)a + xb$ thus $(0,1)$ is path connected thus connected.

- In $(0,1)$, remove x_0 :

$$(0,1) \setminus \{x_0\} = (0, x_0) \cup (x_0, 1).$$

This is the union of two disjoint, nonempty open intervals and is therefore disconnected.

Since f is a homeomorphism we know that its restriction is still a homeomorphism thus $(0,1) \setminus \{x_0\} \cong (0,1] \setminus \{1\}$ However, homeomorphisms must preserve connectedness, f would force a connected space to be homeomorphic to a disconnected one. This contradiction shows that no homeomorphism exists between $(0,1)$ and $(0,1]$.

2. $(0,1)$ vs. $[0,1]$

Assume: There exists a homeomorphism

$$h : (0,1) \rightarrow [0,1].$$

Since 1 is an element of $(0,1)$, there must be some $x_0 \in (0,1)$ with

$$h(x_0) = 1.$$

- In $[0,1]$, remove 0:

$$[0,1] \setminus \{0\} = (0,1],$$

which is connected by same logic as above.

- In $(0,1)$, remove the point x_0 :

$$(0,1) \setminus \{x_0\} = (0,x_0) \cup (x_0,1),$$

which is disconnected.

Again, we have a homeomorphism sending a disconnected space to a connected space, which is impossible.

Thus, no homeomorphism exists between $(0,1)$ and $[0,1]$.

3. $(0,1]$ vs. $[0,1]$

Assume: There exists a homeomorphism

$$g : [0,1] \rightarrow (0,1].$$

Since 0,1 is an element of $[0,1]$, there must be some $x_0, x_1 \in (0,1]$ with

$$g(0) = x_0, \quad g(1) = x_1$$

- In $[0,1]$, remove 0,1:

$$[0,1] \setminus \{0,1\} = (0,1),$$

which is connected by same logic as above.

- In $(0,1]$, remove the point x_0, x_1 :

$$(0,1] \setminus \{x_0, x_1\}$$

If x_0 or $x_1 = 1$ then WLOG let $x_1 = 1$

$$(0,1] \setminus \{x_0, x_1\} = (0,x_0) \cup (x_0,1)$$

If $x_0, x_1 \neq 1$ then WLOG let $x_0 < x_1$

$$(0,1] \setminus \{x_0, x_1\} = (0,x_0) \cup (x_0,x_1) \cup (x_1,1]$$

these are both disconnected.

Again, we have a homeomorphism sending a connected space to a disconnected space, which is impossible.

Thus, no homeomorphism exists between $(0,1]$ and $[0,1]$.

(b) Assume $f : \mathbb{R} \rightarrow \mathbb{R}^n$ is a homeomorphism, thus let $f(0) = x_0$.

Consider the homeomorphism of

$$f|_{\mathbb{R} \setminus \{0\}} : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}^n \setminus \{x_0\}$$

By problem 3 we know that $\mathbb{R}^n \setminus \{x_0\}$ is connected. However, $\mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, \infty)$ thus it is disconnected. Therefore, this is a contradiction since a homeomorphism cannot send a disconnected space to a connected space.

Problem 5

(a) Consider $\text{int}(A) \cap \partial(A) = (X \setminus \overline{X \setminus A}) \cap (\overline{A} \cap \overline{X \setminus A}) = X \setminus \overline{X \setminus A} \cap \overline{A} \cap \overline{X \setminus A} = \emptyset$.
Since you cannot have an element that is in $X \setminus \overline{X \setminus A} \cap \overline{X \setminus A}$.

(b) Consider $\text{int}(A) \cap \partial(A) = (X \setminus \overline{X \setminus A}) \cup (\overline{A} \cap \overline{X \setminus A})$

- If $x \in \text{int}(A)$ since $\text{int}(A)$ is made up of all unions of open sets in A we know that $x \in A \subseteq \bar{A}$.
- If $x \in \partial(A) = \overline{A} \cap \overline{X \setminus A} \implies x \in \bar{A}$
Thus $\text{int}(A) \cup \partial(A) \subseteq \bar{A}$.
- Let $x \in \bar{A}$.
- If $x \notin \overline{X \setminus A}$ then $x \in X \setminus \overline{X \setminus A} \subseteq \text{int}(A) \cup \partial(A)$
- If $x \in \overline{X \setminus A} \implies x \in \overline{X \setminus A} \cup \bar{A} \subseteq \text{int}(A) \cup \partial(A)$

Thus $\text{int}(A) \cup \partial(A) = \bar{A}$

(c) Assume A is open then $\bar{A} \neq A$ and $X \setminus A$ is closed. Thus, $\overline{X \setminus A} = X \setminus A$.

Let $x \in \partial A = \bar{A} \cap (X \setminus A)$. Thus, $x \in \bar{A}$ and $x \in X \setminus A \implies x \notin A \implies x \in \bar{A} \setminus A$.

Let $x \in \bar{A} \setminus A \implies x \in \bar{A}$ and $x \in X \setminus A \implies x \in \bar{A} \cap (X \setminus A) = \partial(A)$

Thus $\partial(A) = \bar{A} \setminus A$ if A is open.

(d) Consider $A = (0,1) \subseteq \mathbb{R}$ which is connected. $\bar{A} = [0,1]$ and $\text{int}(A) = (0,1)$

Thus, $\partial(A) = \{0,1\}$ this is not connected, since consider $(-1/2, 1/2) \cap \{0,1\} = \{0\}$ and $(0, 1.5) \cap \{0,1\} = \{1\}$. These are open in the subspace topology, and therefore $\{1\} \cup \{0\} = \{1,0\}$. Thus not connected.

(e) Consider $A = \mathbb{Q} \cap [0,1]$ this is disconnected since $A = (A \cap (-\infty, 1/2)) \cup (A \cap (1/2, \infty))$.

Assume for contradiction that $x \in \text{int}(A)$ thus there is some U open in $\mathbb{R}$ s.t. $x \in U$. Thus there is an open interval $(-\epsilon + x, x + \epsilon) \subset U$. Thus consider $x - \epsilon < q < x + \epsilon$ for some $q \in \mathbb{Q}$. Fix $N \in \mathbb{N}$ big enough so $\sqrt{2}/n < x + \epsilon - q$. Consider $y = q + \sqrt{2}/n$. This is irrational.

Thus, $x - \epsilon < q < q + \sqrt{2}/n$

Also, $q + \sqrt{2}/n < q + (x + \epsilon - q) = x + \epsilon$.

Thus $y \in (x - \epsilon, x + \epsilon)$. Thus this is a contradiction since we found a irrational number in our interval thus $\text{int}(A)$ is empty.

Since $A \subset [0,1]$ and $[0,1]$ is closed we know that $\bar{A} \subset [0,1]$

Let $x \in [0, 1]$. Let U be an open neighborhood of x . Thus there is an $\epsilon > 0$ s.t. $(x - \epsilon, x + \epsilon) \subset U$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$ we know that there must be a rational number in $(x - \epsilon, x + \epsilon)$. Thus, this interval intersects A thus $x \in \bar{A}$.

Therefore $\bar{A} = [0, 1]$.

Now by (b) we know that $\text{int}(A) \cup \partial(A) = \bar{A}$ thus $\partial(A) = \bar{A} = [0, 1]$. Thus, $\partial(A)$ is connected. Therefore, (e) is false by contrapositive with counter example.

Problem 6

We know from earlier homework that the determinant function

$$\det: GL(n, \mathbb{R}) \rightarrow \mathbb{R}$$

is continuous. Consider the sets

$$\det^{-1}((-\infty, 0)) = \{A \in GL(n, \mathbb{R}) \mid \det(A) < 0\}$$

and

$$\det^{-1}((0, \infty)) = \{A \in GL(n, \mathbb{R}) \mid \det(A) > 0\}.$$

Since $(-\infty, 0)$ and $(0, \infty)$ are open in $\mathbb{R}$ and the determinant is continuous, these preimages are open in $GL(n, \mathbb{R})$.

Moreover, these two sets are disjoint (because no matrix can have both a positive and a negative determinant) and their union is the whole set $GL(n, \mathbb{R})$. Hence, $GL(n, \mathbb{R})$ is not connected.